

1. Name of the medicinal product

MIGRÉTIL, film-coated tablet.

2. Qualitative and quantitative composition

Paracetamol 400 mg; caffeine 100 mg; ergotamine tartrate 1 mg; Belladonna alkaloids 0.1 mg; per tablet.

For excipients, see 6.1.

3. Pharmaceutical Form

Film-coated tablets.

4. Clinical particulars

4.1. Therapeutic indications

Treatment of migraine attacks. It is not indicated as preventive treatment.

4.2. Posology and method of administration

Posology

Except other medical indication, dosage shall be as follows:

Adults: Two tablets taken together, during the aura or at onset of headache. If necessary, another tablet may be administered after half an hour; this dose may also be repeated after another half an hour.

Adolescents: One tablet during the aura or at onset of headache. If necessary, another tablet may be administered after half an hour.

It is not indicated for children under 12 years of age.

Method of administration

Tablets should be taken as soon after the first signs of a migraine attack as possible.

Do not take more than 6 tablets a day or 10-12 tablets a week.

Tablets should be swallowed whole, with some water.

4.3. Contraindications

Hypersensitivity to any of the ingredients. Pregnancy and lactation, active alcoholism, significant hepatic or renal impairment, and septicæmia. Recent or planned angioplasty or vascular surgery, occlusive peripheral vascular disease, uncontrolled hypertension, coronary disease.

Concomitant use of macrolides (namely erythromycin, clarithromycin, and troleandomycin), protease inhibitors (e.g., ritonavir and indinavir), azole antifungals (e.g., ketoconazole and itraconazole), and other drugs that inhibit the CYP3A4 isoenzyme of cytochrome P450 system, should be avoided.

4.4. Special warnings and special precautions for use

The risk of complications increases in cases of severe infection or significant renal impairment, as well as in the elderly. Due to possible reduction of cerebral, cardiac or peripheral blood flow, it should not be the drug of choice in the elderly.

In patients with anxiety disorders, caffeine may exacerbate the symptoms. Concomitant use of drinks containing caffeine should be avoided.

Patients with prostatic hypertrophy, urinary retention, occlusive gastrointestinal disease or glaucoma may undergo deleterious reactions due to belladonna alkaloids, although the doses included in MIGRÉTIL are very low.

The dose shall not be increased or the administration be prolonged because of the risk of ergotism (intense vasoconstriction), gangrene, cardiac vascular fibrosis, retroperitoneal fibrosis or pleuropulmonary fibrosis.

4.5. Interactions with other medicinal products and other forms of interaction

Xanthines may increase the stimulant effects of beta-adrenergics. The concomitant use of central nervous system stimulants may induce additive and undesirable effects, such as nervousness, irritability, insomnia, convulsions and arrhythmias. The concomitant use of monamine oxydase inhibitors may cause tachycardia, arrhythmias or hypertension. Caffeine antagonizes the effects of adenosine and beta-blockers.

The concomitant administration of systemic vasoconstrictors (cocaine, epinephrine, metaraminol, metoxamine, norepinephrine, phenylephrine, local anaesthetic solutions with vasoconstrictors, sympathomimetic amines), as well as smoking, potentiate the vasopressor effects of the ergot alkaloids. These drugs antagonize the nitroglycerine vasodilator effects. Special precautions should be taken with the simultaneous use of propranolol and high doses of ergot alkaloids, due to the risk of severe peripheral vasoconstriction.

The vasoconstrictive effects of ergot alkaloids are additive with those of sumatriptan. Thus, the concomitant use of these drugs shall respect the time interval of at least 24 hours. The administration of ergotamine to individuals with heavy smoking habits may increase the risk of peripheral vascular ischemia, since nicotine has also vasoconstrictor effects.

Concomitant use of sibutramine may increase the risk of hypertension and / or tachycardia.

Excessive alcohol intake may increase the risk of hepatotoxicity induced by paracetamol. Caffeine potentiates the effects of other xanthines, and may reduce their elimination. Macrolides (such as erythromycin, clarithromycin, and troleandomycin), protease inhibitors (e.g., ritonavir and indinavir), azole antifungals (e.g., ketoconazole and itraconazole), and other drugs that inhibit the CYP3A4 isoenzyme of the cytochrome P450 system decrease metabolism of ergot alkaloids, increasing the risk of vasospasm, cerebral ischemia or severe peripheral ischemia.

4.6. Pregnancy and lactation

It should not be used during pregnancy and lactation.

Paracetamol

Epidemiological studies on neurodevelopment in children exposed to paracetamol in utero show inconclusive results.

4.7. Effects on ability to drive and use machines

At recommended doses it does not interfere with the ability to drive or use machines.

4.8. Undesirable effects

At recommended doses, MIGRÉTIL is usually well tolerated.

Notwithstanding the following side effects may occur:

Gastrointestinal disorders: signs of gastric irritation, nausea, vomiting or diarrhea.

Nervous system disorders: insomnia, nervousness, vertigo, myalgia, paresthesia and decrease of the lower limbs strength.

Cardiac disorders: The effects of vasoconstriction may be felt in any vessel, with great individual variability. These effects appear as cold, numb and painful extremities, with or without paresthesias.

Other symptoms such as absence of pulse, bradycardia, cardiac valvular fibrosis, cyanosis, edema, changes in the electrocardiogram, hypertension, ischemia, precordialgia, tachycardia, and vasospasm are also described.

Immune system disorders: skin hypersensitivity reactions, such as itching.

Others: processes of fibrosis, besides cardiac valves, may reach the retroperitoneal space and the pulmonary interstitium.

4.9 Overdose

Cases of excessive dosage or acute intoxication may occasionally occur in patients who take doses higher than the recommended or in subjects with suicidal tendencies. Symptoms may include nausea, vomiting, abdominal pain, headaches, excitability, agitation, scotoma, tinnitus, tremors, tachycardia, arrhythmias, diuresis increase, myalgia, paresthesia, weakness of legs, cold, numb and painful extremities. In severe cases angor, transient monocular blindness, bilateral papillitis, ischemic colitis, renal artery vasoconstriction, gangrene of the extremities, hepatic necrosis, mental disturbance, delirium, dyspnea, hypotension or hypertension, weak pulse, muscular spasms, seizures, shock and death.

Treatment of severe overdose consists of general supportive measures and symptomatic therapy. Induction of emesis and gastric lavage are important if the intake was less than 4 hours before, as well as the administration of activated charcoal. The use of demulcents may reduce gastric irritation. Extremities should be kept warm and eventually vasodilators and heparine may be administered. In the event of hepatic necrosis, secondary to paracetamol, acetylcysteine should be given within the first 24-36 hours, since it is more effective the earlier it is administered.

5. PHARMACOLOGICAL PROPERTIES

5.1. Pharmacodynamic properties

Pharmacotherapeutic group: II - 11 – Medicines used to treat migraines.

ATC Classification: N02CA – Antimigraine preparations, ergotamine alkaloids

Paracetamol: It produces analgesic and antipyretic effects by a mechanism not completely understood, but which may be similar to that of salicylates. In terms of analgesia, paracetamol may act by inhibiting the synthesis of prostaglandins in the central nervous system and, at a lower level, blocking the genesis of the impulse of pain, peripherically. Peripheral action may also be attributable to the inhibition of prostaglandins or to the

inhibition of synthesis and/or the action of other substances that make the pain receptors sensitive to mechanical or chemical stimulation.

Caffeine: It is a methylxanthine, occurring naturally in tea and coffee, but it is prepared synthetically for commercial use. It potentiates the action of ergot alkaloids in the management of migraine, increasing ergotamine gastrointestinal absorption.

Caffeine may also contribute to cerebral vasoconstriction and the consequent reduction of blood flow and local oxygen tension. It has been used in association with analgesics, such as paracetamol, in the management of headache. Caffeine does not exert any intrinsic analgesic effect; however, preparations containing caffeine have been described as having a faster and more potent analgesic effect, thus allowing the doses of the analgesic substance to be reduced.

Ergotamine tartrate: Ergotamine is an ergot alkaloid that keeps being an important agent in the symptomatic relief of cephalgia of migraine, particularly in patients for whom aspirin-like drugs only give partial relief. It has complex pharmacological effects. In therapeutic doses it causes peripheral vasoconstriction, stimulating the alfa-adrenergic receptors and suppressing the central vasomotor centres.

The mechanism by which ergotamine acts on vascular headaches is probably due to direct vasoconstriction of the external carotid artery, which leads to the decrease in the pulse amplitude. The drug's effects on catecholamines and serotonin are also involved in its therapeutic efficacy.

Belladonna alkaloids: They are anticholinergic drugs. However, the doses used in MIGRÉTIL are not enough to produce such effect. Gastric stasis, nausea and vomiting are symptoms and signs present in most cases of migraine. Belladonna alkaloids exert a beneficial anti-emetic effect, contributing to the symptomatic relief of the attack.

5.2. Pharmacokinetic properties

Paracetamol: It is rapidly and almost completely absorbed from the GI tract. Peak plasma concentration is attained within 10-60 minutes. Following a 500 mg-dose, peak concentration is 2.1 mcg/mL. It is rapidly and uniformly distributed into most tissues including breast milk. It crosses the placenta. About 25% of paracetamol in blood is bound to plasma proteins. Plasma half-life is from 1.25-3 hours. It is metabolized by microsomal enzyme systems in the liver. About 80-85% of the administered dose undergoes conjugation principally with glucuronic acid and to a lesser extent with sulfuric acid. A small amount is conjugated with cysteine. Paracetamol is excreted by urinary route, free or as conjugate. Approximately 85% of the dose is excreted in the urine within 24 hours after a single dose administration.

Caffeine: Caffeine is well absorbed from the GI tract, with oral absorption being higher than absorption via intramuscular injection. Following oral administration of 100 mg of caffeine, peak plasma concentrations of about 1.5-1.8 mcg/ml are reached after 50-75 minutes. Distribution is wide and rapid, crossing the blood-brain barrier and the placenta, and is distributed into breast milk. Approximately 17% of the drug is bound to plasma proteins. Caffeine is partially demethylated, oxydated and acetylated in the liver, and its metabolites are excreted in urine. Only 1% is excreted unchanged. It has a plasma half-life of 3-4 hours in adults, and it can be more than 100 hours in the newborn.

Ergotamine tartrate: Following oral administration, absorption of ergotamine tartrate varies. It is reduced by the gastric stasis normally accompanying migraine. Absorption is greater within the first hour after oral administration, particularly when the drug is taken with caffeine.

The first-pass effect helps to reduce its bioavailability. It crosses the blood-brain barrier and is detected in breast milk. About 95% circulates bound to plasma proteins. Peak plasma concentration is 2 hours, decreasing to half when the drug is administered simultaneously with caffeine. The onset of action depends on how promptly the drug is given after the onset of the headache. Following oral administration of ergotamine in individuals with normal renal and hepatic function, plasma concentrations declined biphasically, with an average half-life of 2.7 hours in the initial phase and 21 hours in the terminal phase. Ergotamine, particularly the peptide portion of the molecule, is extensively metabolized, probably in the liver. Some of its metabolites are pharmacologically active. The principal route of excretion is the feces and only about 4% of the dose is excreted in urine.

Belladonna alkaloids: They are well absorbed from the GI tract. Little information is available on their distribution, metabolism and excretion. They are probably hydrolyzed to tropine and tropic acid, the derivatives of which are then excreted in the urine, and to a lesser extent, in feces.

5.3. Preclinical safety data

MIGRÉTIL compounds were not subjected to mutagenicity and carcinogenicity studies; however, there are no reports concerning any of these effects in humans, despite the long use in clinical practice of these compounds, alone or combined.

Paracetamol

Conventional studies using the currently accepted standards for the evaluation of toxicity to reproduction and development are not available.

Ergotamine

Ergotamine may cause abortion and fetal malformations, so its use during pregnancy should be avoided.

6. PHARMACEUTICAL PARTICULARS

6.1. List of excipients

Povidone, tartaric acid, microcrystalline cellulose, sodium croscarmellose, magnesium stearate, Eudragit E100, hydroxypropylmethylcellulose, diethyl phthalate, and Opaspray K-1-7000.

6.2. Incompatibilities

Atropine sulphate is incompatible with alkalis, tannic acid and mercury salts.

Scopolamine hydrobromide is incompatible with alkalis, silver salts and tannic acid.

6.3. Shelf-life

Two years.

6.4. Special precautions for storage

Store in the original package. Do not store above 30°C.

6.5. Nature and contents of container

Tablets are contained in ALU/ALU in cardboard boxes containing 20 or 60 tablets. Not all pack sizes may be marketed.

6.6. Instructions for use and handling

No special requirements.

7. MARKETING AUTHORIZATION HOLDER

BIAL – Portela & C^a, S.A.
À Av. da Siderurgia Nacional
4745-457 S. Mamede do Coronado
Portugal

8. MARKETING AUTHORIZATION NUMBER

9. DATE OF FIRST MARKETING AUTHORIZATION

10. DATE OF REVISION OF TEXT